

Generation Interconnection Request System Impact Study Report

For Queue Project Q17-0848

Kickapoo 100 kV
February 22, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0848. The Interconnection Customer proposes a 215.1 MW solar photovoltaic facility interconnecting at the Kickapoo 100 kV substation in the Ohio area. The request was received on August 14, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 317 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$49,207,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Stryker – Hesperia 100 kV Line, the Lake Marion – San Andres Substation 100 kV loads to 215.4% of its 164 MVA rating. Project DFAX: 100.0%.

San Andres – Bark Riv 100 kV: 215.4% loading (164 MVA) following the outage of the Hesperia – Stryker 100 kV; DFAX = 100.0%.

Hesperia – Chub Lake 100 kV: 208.3% loading (131 MVA) following the outage of the E Batavia – Kickapoo 100 kV L; DFAX = 69.1%.

The Covanta Hennepin Energy – Chub Lake Substation 100 kV Line is overloaded to 204.8% of its 125 MVA emergency rating for the loss of the Kickapoo – E Batavia Substation 100 kV. The project contributes 69.1% DFAX (148.6 MW) to this overload.

For the loss of Hesperia – Strker 100 kV, the Bark Riv – E Batavia 100 kV loads to 202.2% of its 185 MVA rating. Project DFAX: 100.0%.

Kickapoo – Stryker 100 kV: 191.7% loading (205 MVA) following the outage of the Kickapoo – E Batavia 100 kV; DFAX = 100.0%.

For the loss of Kickapoo – E Batavia 100 kV, the Hesperia – Stryker 100 kV loads to 191.7% of its 205 MVA rating. Project DFAX: 100.0%.

The E Batavia – Kickapoo 100 kV is overloaded to 191.7% of its 205 MVA emergency rating for the loss of the Stryker – Hesperia 100 kV. The project contributes 100.0% DFAX (215.1 MW) to this overload.

For the loss of E Batavia – Kickapoo 100 kV, the Lakeside 2 – Covanta Hennepin Energy 100 kV loads to 184.9% of its 131 MVA rating. Project DFAX: 69.1%.

For the loss of Strykr – Hesperia 100 kV, the Lake Maron – Pleasant Street 100 kV loads to 148.2% of its 112 MVA rating. Project DFAX: 31.8%.

The Pleasant St – Valley Children'S Hospital Substation 100 kV is overloaded to 136.0% of its 118 MVA emergency rating for the loss of the Hesperia – Stryker 100 kV. The project contributes 31.8% DFAX (68.4 MW) to this overload.

For the loss of Chub Lake Substation – Hesperia 100 kV line, the Hesperia – Border Tap 100 kV line loads to 123.2% of its 105 MVA rating. Project DFAX: 30.6%.

The 96Th St – Vly Children'S Hospital 100 kV L is overloaded to 122.7% of its 124 MVA emergency rating for the loss of the Hesperia – Stryer 100 kV. The project contributes 31.8% DFAX (68.4 MW) to this overload.

Lakeside 2 Substation – Merced Falls 100 kV: 118.6% loading (138 MVA) following the outage of the Landsman Crk – Russellville 100 kV; DFAX = 5.2%.

Merced Falls – Manley Sand 100 kV L: 113.7% loading (136 MVA) following the outage of the Landsman Creek – Russellville 100 kV; DFAX = 5.2%.

Mayberry – City Of Bryan 100 kV line: 113.2% loading (109 MVA) following the outage of the Hesperia – Chub Lake Substation 100 kV Line; DFAX = 30.6%.

The Chouteau – 96Th St 100 kV is overloaded to 110.6% of its 129 MVA emergency rating for the loss of the Hesperia Substation – Stryker 100 kV. The project contributes 31.8% DFAX (68.4 MW) to this overload.

The Border Tap Substation – Mayberry 100 kV is overloaded to 109.4% of its 113 MVA emergency rating for the loss of the Hesperia – Chub Lake 100 kV. The project contributes 30.6% DFAX (65.8 MW) to this overload.

For the loss of Landsman Crk – Russellville 100 kV, the Lost Riv – Manley Sand Substation 100 kV L loads to 107.5% of its 132 MVA rating. Project DFAX: 5.2%.

The Monroeville – Woodburn 100 kV is overloaded to 107.2% of its 144 MVA emergency rating for the loss of the Hesperia – Stryker 100 kV. The project contributes 8.7% DFAX (18.8 MW) to this overload.

The Sage – Woodburn 100 kV is overloaded to 106.9% of its 155 MVA emergency rating for the loss of the Hesperia – Stryker 100 kV Line. The project contributes 8.7% DFAX (18.8 MW) to this overload.

For the loss of Hesperia – Stryker 100 kV, the Monte Del Estado 6303 – Sage 100 kV line loads to 106.7% of its 171 MVA rating. Project DFAX: 8.7%.

For the loss of Swan Vly – Muskogee 500 kV, the Dean Forest – Eastown Road 230 kV loads to 106.3% of its 643 MVA rating. Project DFAX: 6.2%.

Dean Forest – Muskogee 230 kV: 105.8% loading (659 MVA) following the outage of the Muskogee – Swan Vly 500 kV L; DFAX = 6.2%.

The Lost River – Cajalco 100 kV is overloaded to 103.1% of its 127 MVA emergency rating for the loss of the Russellville – Landsma Creek 100 kV. The project contributes 5.2% DFAX (11.2 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lake Marion – San Andres Substation 100 kV	Rebuild line with higher-capacity conductor	\$4,475,000	54% (\$2,411,000)
San Andres – Bark Riv 100 kV	Rebuild line with higher-capacity conductor	\$2,371,000	54% (\$1,277,000)
Hesperia Substation – Chub Lake 100 kV	Rebuild line with higher-capacity conductor	\$4,339,000	52% (\$2,273,000)
Covanta Hennepin Energy – Chub Lake Substation 100 kV Line	Rebuild line with higher-capacity conductor	\$3,758,000	52% (\$1,969,000)
Bark Riv – E Batavia 100 kV	Rebuild line with higher-capacity conductor	\$4,443,000	54% (\$2,394,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Kickapoo – Stryker 100 kV	Rebuild line with higher-capacity conductor	\$929,000	54% (\$501,000)
Hesperia Substation – Stryker 100 kV Line	Rebuild line with higher-capacity conductor	\$7,739,000	54% (\$4,170,000)
E Batavia – Kickapoo 100 kV	Rebuild line with higher-capacity conductor	\$3,725,000	54% (\$2,007,000)
Lakeside 2 – Covanta Hennepin Energy 100 kV	Rebuild line with higher-capacity conductor	\$4,557,000	52% (\$2,388,000)
Pleasant St – Lake Marion 100 kV line	Rebuild line with higher-capacity conductor	\$11,910,000	54% (\$6,417,000)
Pleasant St – Vly Children'S Hospital 100 kV	Rebuild line with higher-capacity conductor	\$6,822,000	54% (\$3,675,000)
Hesperia – Border Tap 100 kV line	Rebuild line with higher-capacity conductor	\$8,329,000	54% (\$4,488,000)
96Th St – Vly Children'S Hospital 100 kV L	Rebuild line with higher-capacity conductor	\$4,743,000	54% (\$2,555,000)
Lakeside 2 Substation – Merced Falls 100 kV	Rebuild line with higher-capacity conductor	\$5,882,000	37% (\$2,170,000)
Merced Falls – Manley Sand 100 kV L	Rebuild line with higher-capacity conductor	\$5,579,000	37% (\$2,059,000)
Mayberry – City Of Bryan 100 kV line	Rebuild line with higher-capacity conductor	\$415,000	54% (\$224,000)
96Th St Substation – Chouteau 100 kV	Rebuild line with higher-capacity conductor	\$3,846,000	54% (\$2,072,000)
Border Tap Substation – Mayberry 100 kV	Reconductor line	\$3,265,000	54% (\$1,759,000)
Manley Sand – Lost Riv 100 kV	Reconductor line	\$853,000	37% (\$315,000)
Monroeville – Woodburn 100 kV	Reconductor line	\$3,739,000	39% (\$1,444,000)
Sage – Woodburn 100 kV	Reconductor line	\$993,000	39% (\$383,000)
Monte Del Estado 6303 – Sage Substation 100 kV Line	Reconductor line	\$976,000	39% (\$377,000)
Eastown Road – Dean Forest 230 kV	Reconductor line	\$9,603,000	3% (\$306,000)
Muskogee – Dean Forest 230 kV Line	Reconductor line	\$18,726,000	3% (\$597,000)
Lost River – Cajalco 100 kV	Reconductor line	\$2,646,000	37% (\$976,000)
	Total allocated to project		\$49,207,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.